

## **Finals Essay Task 1.**

Assessment on your Understanding as to what really matters to become Data Analysts.

And How to integrate SQL into your Data Analysis

**1. What are the four primary factors that will differentiate a data analyst candidate in 2026?**

- The four primary factors are domain knowledge, understanding the technical ecosystem, soft skills that matter (good communication skills) and strong reasons why (clear purpose/motivation). These can help a person or stand out, it shows that they can work well and be consistent.

**2. How does domain knowledge provide a competitive advantage over purely technical candidates?**

- It helps because you already understand the metrics, customers, stakeholders and the company dynamics. These are some of the things that a purely technical candidate cannot compete with. And, this makes it easier to explain data and give useful things.

**3. What is the "lean tech stack" recommended for a foundational data analyst role?**

- The basic tools that are recommended are Excel (1-3 weeks), SQL (3-6 weeks) to be able to write queries with multiple joins, and Visualization tools like the tableau or Power BI (1-3 weeks). These are enough to start analyzing and showing the data in a clear way.

**4. Distinguish between "learning projects" and "showing projects."**

- Learning projects are low pressure practice and used for practice with one tool at a time using clean datasets. While, showing projects are more advanced and involved solving real business questions using multiple tools and messy data. But they both show your skills and how you explain your results but are different in level of difficulty.

**5. According to the roadmap, why is Python often deprioritized for mid-career data analyst positions?**

- Python is often prioritized because most of the time it is not always used in daily work and not often tested in the interview process. Also, most stakeholders cannot understand python code or outputs.

#### **Questions on SQL-Base on the Article Posted**

**6. Why is establishing the "Business Context" essential for an analyst? What is a Business Context?**

- Business context means understanding the data represents, what problem it is trying to solve and who will use or benefit from the output of analysis. It is essential because it helps the analyst know what to analyze.

**7. According to the text, how does SQL function as a "flashlight"?**

- It is a flashlight because it helps you explore step by step and see what is inside the data. It helps you find errors such as duplicate, invalid and negative values.

**8. Why is the validation step considered a professional necessity rather than an optional task?**

- Validation is more necessary because it checks if the data is correct before analyzing it. It prevents mistakes and makes sure that the results are more reliable and credible.

**9. What defines "good SQL" according to the author's standards?**

- Good SQL is easy to read, explain and trust. It should be easy to explain to others. It is not about writing long or complicated queries.

**10. Describe the fundamental mindset shift required to move from being a student to being an analyst.**

- The mindset change is from just writing queries to answering real business questions. A data analyst focuses on the purpose of the query, not the syntax.